

Software Requirements Specification

for

Campus Marketplace Application

Version 0.1

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1. Introduction

1.1 Document Purpose

The purpose of this Requirements Analysis Document (RAD) is to identify, analyze and document the function and non-functional requirements for the Campus Marketplace Application. This document introduces and is a communication tool between stakeholders and developers. It provides a clear understanding of the system requirements and expected functionality.

This document also shows the results of requirements determination and requirements structuring activities that were conducted during the analysis phase of Systems Development Life Cycle (SDLC). The information in the document will help guide the design, development, testing, and implementation of the system.

1.2 Product Scope

The Campus Marketplace App (CMA) is a web-based platform designed for members of the Kwantlen Polytechnic University (KPU) community. The system provides a secure and safe online marketplace where students, faculty, and staff can buy and sell items.

The application enables users to:

- Register and log in using only their KPU credentials.
- Create, edit, and manage product listings.
- Search and browse available items.
- Communicate with buyers and sellers through an internal messaging system.

The system aims to improve accessibility, convenience, and safety compared to public online marketplaces by restricting access to only verified KPU community members.

1.3 Intended Audience and Document Overview

This document is intended for the following audience:

Table 1: Intended Audience

Audience	Purpose
Project Team	To understand system requirements and guide development activities.
Stakeholders	To verify that system requirements accurately reflect user needs.
Future Developers and Maintainers	To provide reference information for future system enhancements and maintenance.
Instructor	To evaluate project progress and requirements analysis activities

This document provides an overview of the Campus Marketplace Application and presents the results of the requirements analysis process. It also includes requirements for determination of findings, system models, functional requirements, non-functional requirements, and supporting documents.

1.4 Document Conventions

The following conventions are used throughout this document:

- Headings are numbered according to hierarchy
- All text is formatted using Times New Roman
- Body text uses 12-point font size
- Level 1 heading use 18-point font size
- Level 2 heading use 14-point font size
- Tables and figures are numbered sequentially throughout the document
- Use cases are labeled UC-x

2. Overall Description

2.1 Product Overview

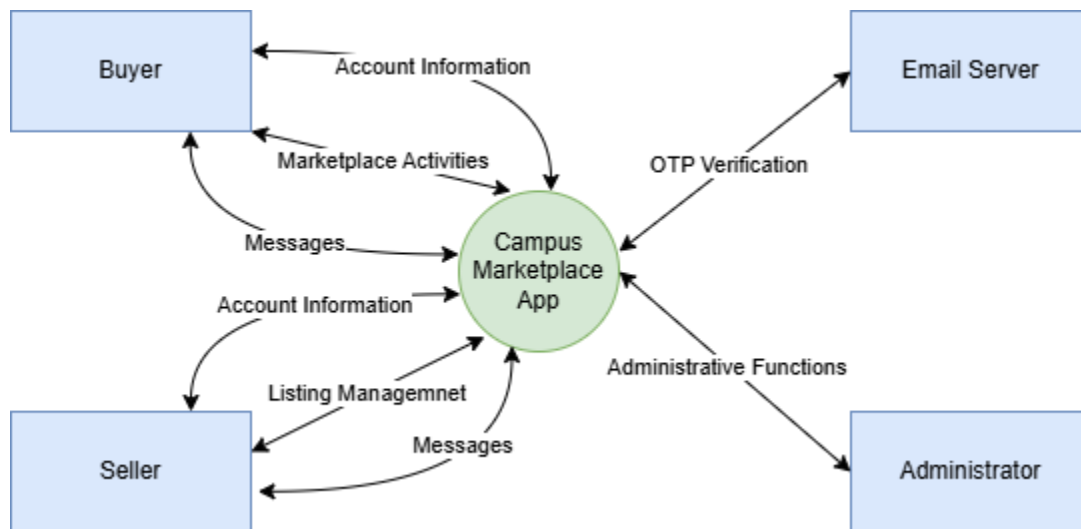
The Campus Marketplace Application is a web-based marketplace designed for the members of Kwantlen Polytechnic University (KPU) community. The application provides a centralized platform where the KPU community members can buy and sell items in a secure and trusted environment.

Currently, many students rely on public marketplace platforms such as Facebook Marketplace or Craigslist to buy and sell items. These platforms often expose users to risks such as frauds, unreliable sellers, and scary experiences. The Campus Marketplace Application addresses these concerns by restricting access to only verified KPU members.

The system allows users to create and manage listings, search for products, and communicate with buyers and sellers through a single platform.

2.1.1 Context Diagram

Figure 2.1.1: Context Diagram



The Context Diagram provides a high-level view of the Campus Marketplace Application and its interactions with external entities. It explains the flow of information between users and the system without showing internal processes.

To reduce the complexity of the diagram we have

External Entities

- Buyer
 - Searches and views listings
 - Contact sellers
 - Receives listing information and messages
- Seller

- Creates, edits, and deletes listings
 - Responds to buyer inquiries
 - Receive notifications and messages
- Administrator
 - Monitors user activity
 - Manages user accounts
 - Observes listings

System

- Campus Marketplace Application

Data Flows

- Account Information
 - Register / Login
- Marketplace Activities
 - Search Listings
 - View Listings
 - Create Listings
 - Edit / Delete Listings
- Messages
 - Send Messages
 - Receive Messages
 - Send Reply
 - Receive Reply
- Administrative Functions
 - Manage Users
 - Moderate Listings

2.2 Product Functionality

User Account Management

- Users register and log in using valid KPU email address
- Users can manage their profiles
- Reset forgotten password

Listing Management

- Users can create, edit, and delete product listings
- Users can upload product images
- Listings will include product description, price, and category tags
- Listing can be marked as sold

Search and Browse

- Users can browse through listings once logged in
- Users can search and filter listings by tags, keywords, or price
- View detailed product information

Communication

- Buyers can send direct messages to sellers through the platform
- Users can receive notifications of messages or product listings

Administration

- Administrators will manage user accounts and listings
- Administrators have control over everything

2.3 Design and Implementation Constraints

Technology Constraints

- The system will be developed using web technologies such as HTML, CSS, JavaScript, and a backend framework
- A relational database will be used for data storage

Security Constraints

- Users must authenticate before posting listings
- User passwords must be stored securely
- Access to administrative functions must be restricted to authorized administrators

Academic Constraints

- The project must be completed within the course timeline
- Development will be performed by a student team

Platform Constraints

- The application must support modern web browsers
- Internet access is required
- The system should be accessible from desktops and mobile devices

2.4 Assumptions and Dependencies**Assumptions**

- Users have access to the internet
- Users possess a valid campus email account
- Users provide accurate information when creating accounts
- Users are responsible for arranging item exchanges

Dependencies

- The application depends on database services for data storage
- The application depends on email services for notifications and account verification
- The application depends on the hosting environment being available
- The application depends on users maintaining active internet connections

3. Requirements Determination**3.1 Interview Summary**

The requirements for the Campus Marketplace Application were gathered through surveys, informal interviews, and discussions with KPU community members and analysis of existing online marketplace platforms. These activities were conducted to better understand the needs and expectations of potential users.

A survey was developed and distributed to KPU members (students, faculty, and staff) to collect feedback regarding their experiences with existing marketplaces and the features they would like to see in a campus-specific marketplace. In addition, informal discussions were held with classmates and peers to gain further insight into shared challenges that are associated with buying and selling items.

The information collected was analyzed and categorized into functional requirements, non-functional requirements, and stakeholder concerns that would guide the development of the project.

3.2 Stakeholder Requirements

The survey responses and dissuasions identified several key requirements and expectations for the Campus Marketplace Application.

3.2.1 User Interface and Design

Participants emphasized the importance of a modern and visually appealing interface. Users indicated that the system should:

- Provide an easy-to-use interface for creating, editing, and managing listings.
- Includes user-friendly search, filtering, and communication features.
- Offer notifications and the ability to save searches or listings for future references.
- Maintain a clean, organized, and uncluttered layout that is easy to navigate.
- Incorporate aesthetically pleasing visuals and a professional graphical user interface (GUI).
- Follow design principles similar to existing marketplace platforms such as Facebook Marketplace, to provide a familiar but better user experience.

3.2.2 Safety and Security

Safety and trust were among the most frequently discussed concerns. Participants suggested that the system should:

- Verify the identity of buyers and sellers using KPU credentials.
- Reduce opportunities for fraud and fraudulent listings.
- Promote safe transaction practices.
- Provide a trusted environment limited to members of the KPU community.

3.2.3 Marketplace Features

Participants identified several desired marketplace features including:

- Buying and selling functionality
- Direct messaging between buyers and sellers
- Simple search and browsing capabilities
- Support for free item listings
- Reduce External Advertisements

3.2.4 Community and Connectivity

Participants communicated interest in strengthening connections within the KPU community through the marketplace application. A campus specific marketplace application would provide a convenient way for students, faculty and staff to exchange goods, services, professional networks and become more connected within a trusted environment.

3.2.5 Potential Concerns

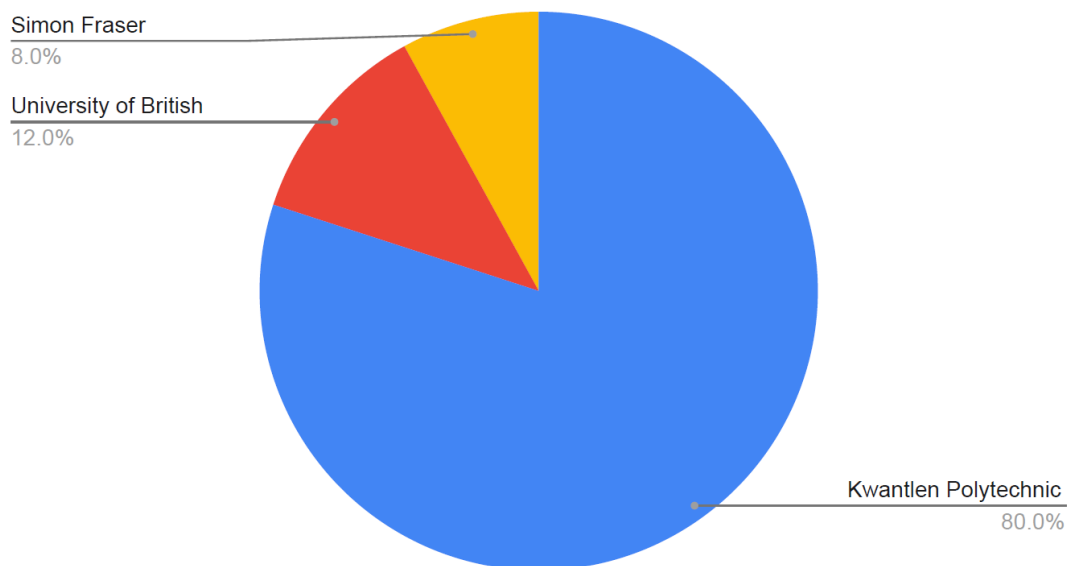
Participants also identified several risks and concerns that should be addressed by the system:

- Professionalism and Community Conduct
 - Maintaining professional boundaries within a shared platform that includes students, faculty members, and staff.
- Platform Misuse
 - Preventing misuse of the platform through inappropriate content, spam, or fraudulent activity.

3.3 Findings and Analysis

Figure 3.3.1: Respondents' University Institution

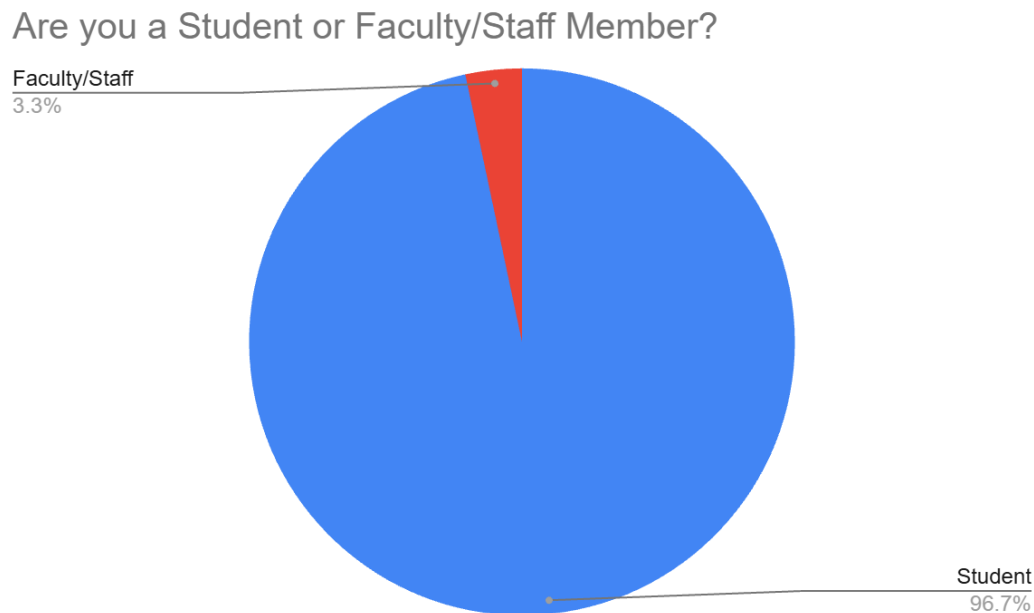
Which University/College are you from?



The majority of respondents (80%) were from Kwantlen Polytechnic University, while 12% were from University of British Columbia and 8% were from Simon Fraser University. This

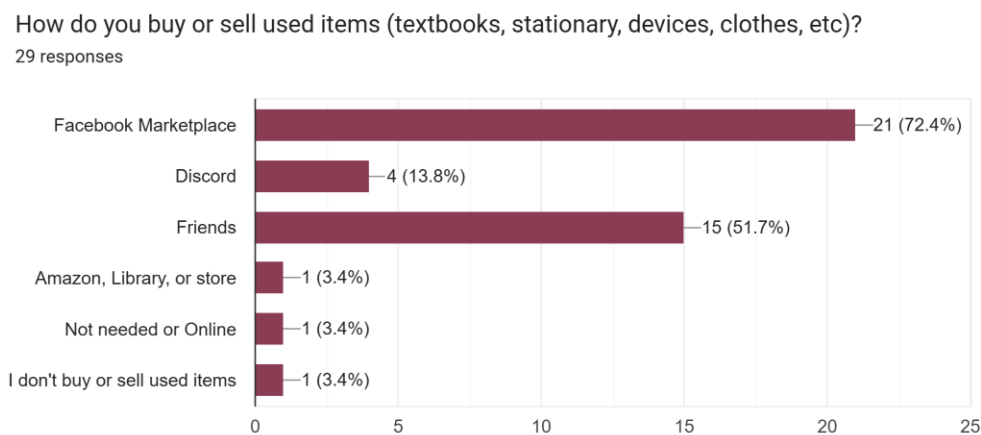
indicates that the survey results primarily reflect the perspectives of the KPU students.

Figure 3.3.2: Respondents Status (Student or Faculty/Staff)



Most respondents (96.7%) identified as students, while 3.3% identified as faculty or staff members. Therefore, the findings largely represent student experiences and perspectives of KPU students.

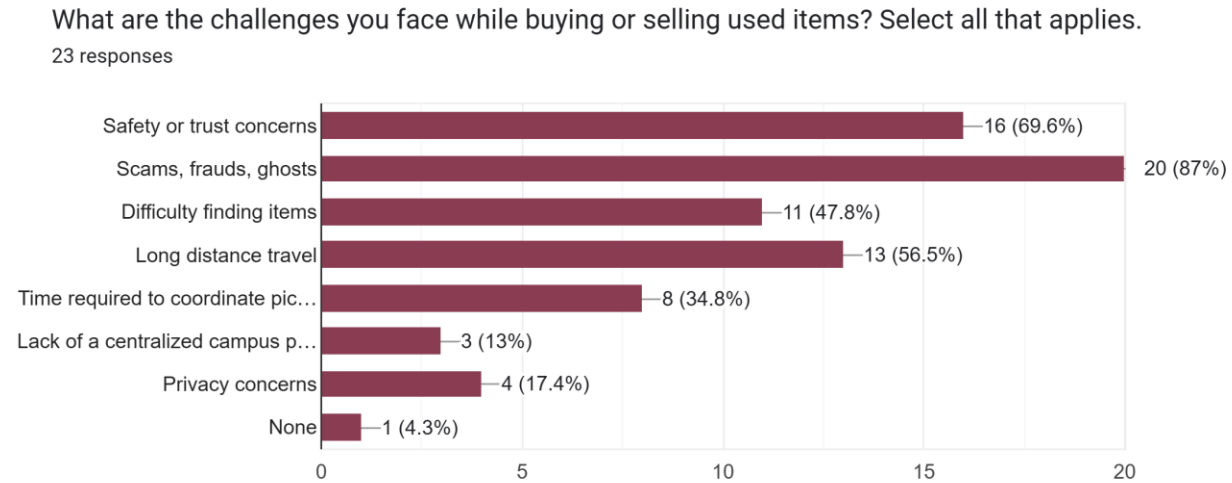
Figure 3.3.3: Platforms Used to Buy or Sell Used Items



Facebook Marketplace was the most used platform (72.4%), followed by transactions through

friends (51.7%). Other channels such as Discord, stores and libraries were used less frequently, indicating a strong reliance on social and online networks.

Figure 3.3.4: Challenges Faced When Buying or Selling Used Items

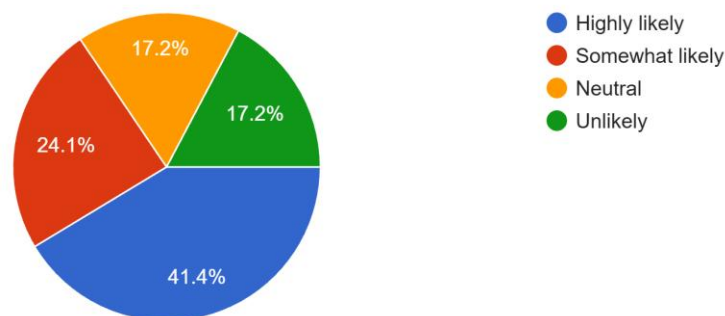


Scam and fraudulent activities were identified as the most significant challenge (87%), followed by safety concerns (69.6%) and long-distance travel (56.5%). These findings highlight the need for a safer and more convenient marketplace solution.

Figure 3.3.5: Likelihood of Using a University Marketplace App

How likely would you use a Marketplace App catered to your university/college?

29 responses

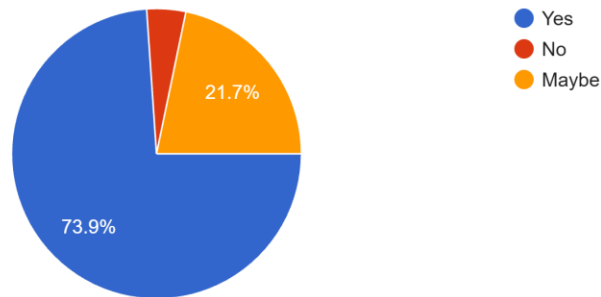


More than 65% of respondents indicated they were highly likely or somewhat likely to use a university-specific marketplace application. This suggests a strong potential demand for a platform like this among our community.

Figure 3.3.6: Impact of University Email Verification on Trust

Users will be verified by their university email - Would this email verification marketplace increase your trust in this platform and feeling less exposed to scammers/frauds?

23 responses

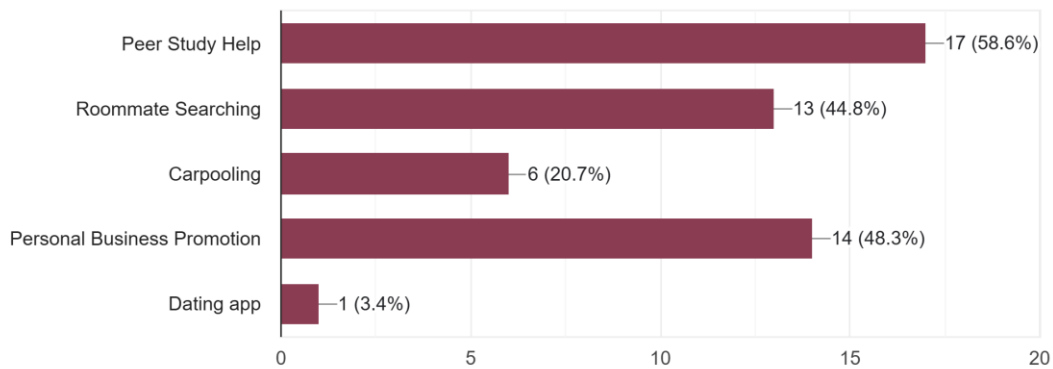


Most respondents (73.9%) believed that university email verification would increase their trust in the platform, while 21.7% were uncertain. This demonstrates that identity verification is an important feature for improving user confidence.

Figure 3.3.7: Interest in Additional Services Beyond Buying and Selling

Beyond physical items, would you use this app for other services? Select all that applies.

29 responses

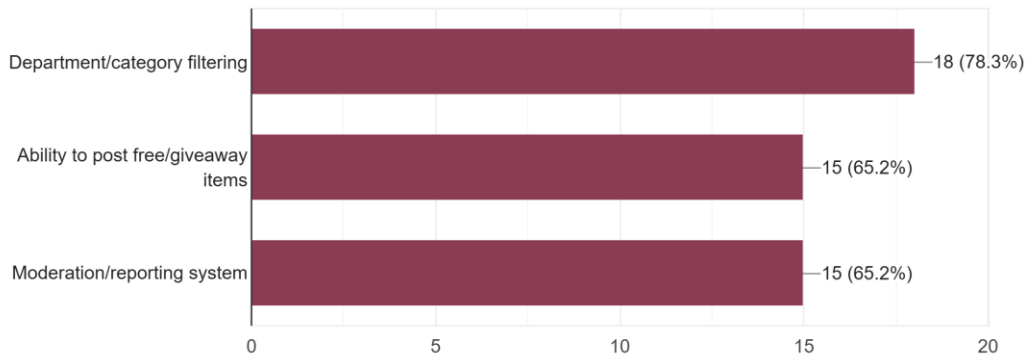


Respondents showed interest in using the platform for additional services such as peer study help (58.6%), personal business promotion (48.3%), and roommate searching (44.8%). This suggests opportunities to expand the platform beyond just a traditional marketplace.

Figure 3.3.8: Features that Would Encourage Platform Usage

What features would make you more likely to use this platform? (Select all that apply)

23 responses

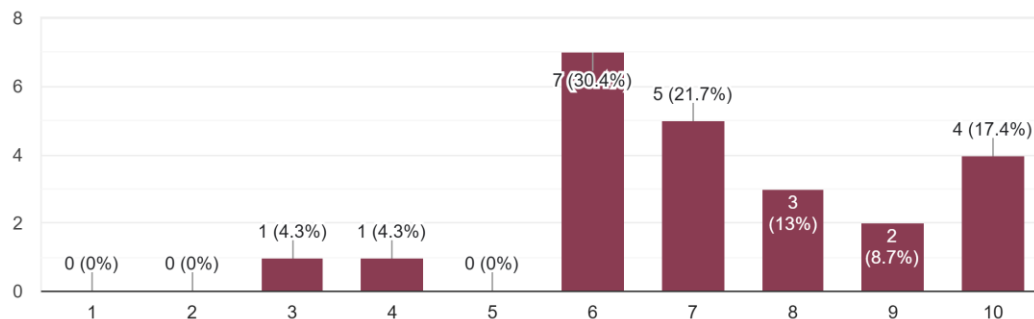


Department or category filtering was the most desired feature (78.3%), followed by free-item postings and moderation/reporting systems (65.2% each). These features will significantly improve usability and user satisfaction.

Figure 3.3.9: User Experiences with Existing Marketplace Platforms

What have been your experiences with other marketplace apps/forums?

23 responses

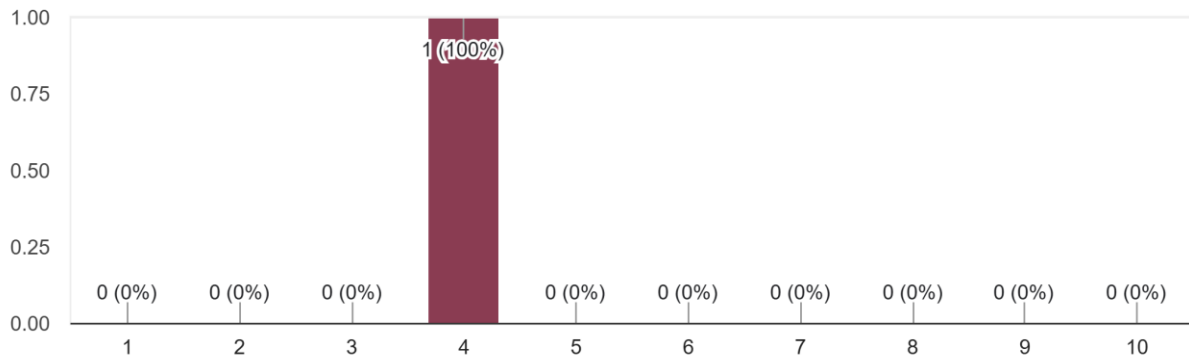


Most respondents rated their experiences with existing marketplace platforms between 6 and 10 on a 10-point scale. The most common rating is 6 out of 10. This indicates generally positive experiences, however with room to improve.

Figure 3.3.10: Additional Experience Rating Response

What have been your experiences with other marketplace apps/forums?

1 response

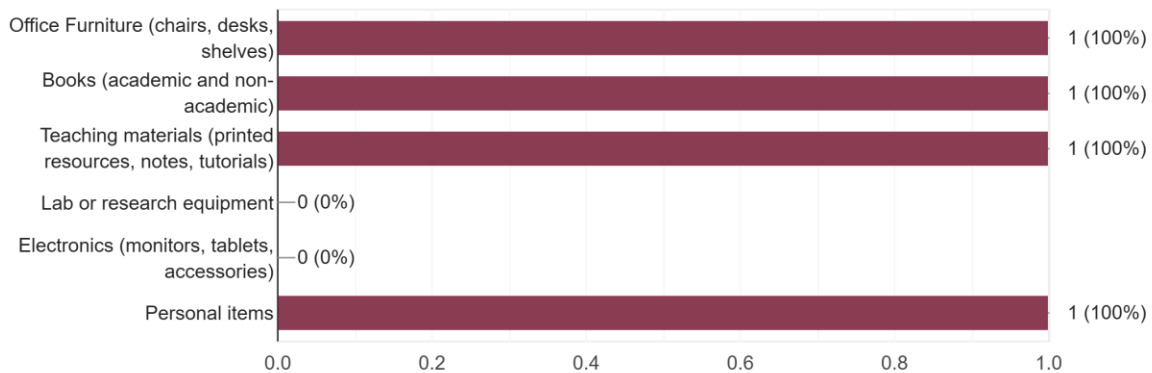


The additional selection provided a rating of 4 out of 10 for existing marketplace experiences. Although based on a limited sample, this suggests that some users have encountered challenges when using current marketplace platforms.

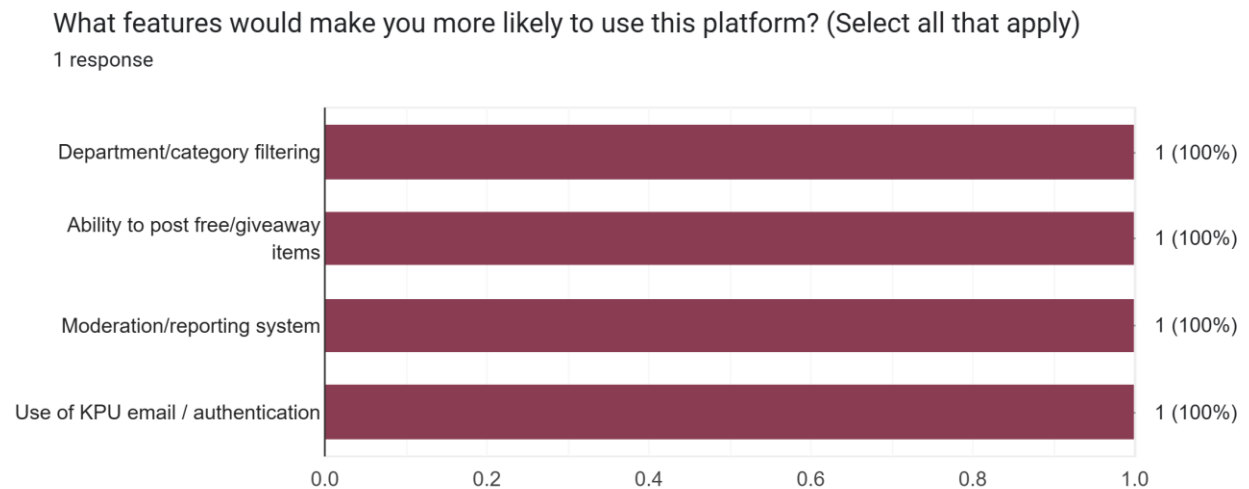
Figure 3.3.11: Types of Items Users Would Most Likely List or Share

What types of items would you be most likely to list or share? (Select all that apply)

1 response



Respondents indicated interest in listing a variety of items, including books, teaching material, furniture, and personal items. This demonstrates the potential diversity of products that could be exchanged through the platform.

Figure 3.3.12: Preferred Platform Features

The respondents selected department/category filtering, free-item posting, moderation/tools, and university email authentication as important features. These responses reinforce the importance of organization, security, and accessibility.

4. Specific Requirements

4.1 External Interface Requirements

The Campus Marketplace Application will provide a web-based interface accessible through modern browsers. The system will support interactions between users and the application through graphical user interfaces (GUI) components.

4.1.1 User Interface Requirements

- The system will provide a responsive web interface accessible on desktop and mobile devices.
- The system will provide forms for user registration, login, and profile management.
- The system will provide interfaces for creating, editing, and viewing listings.
- The system will provide a messaging interface for communication between users.

4.1.2 Hardware Interface Requirements

- The system will operate on devices such as laptops, desktops, tablets, and smartphones.
- An internet connection is required for all system functionalities.

4.1.3 Software Interface Requirements

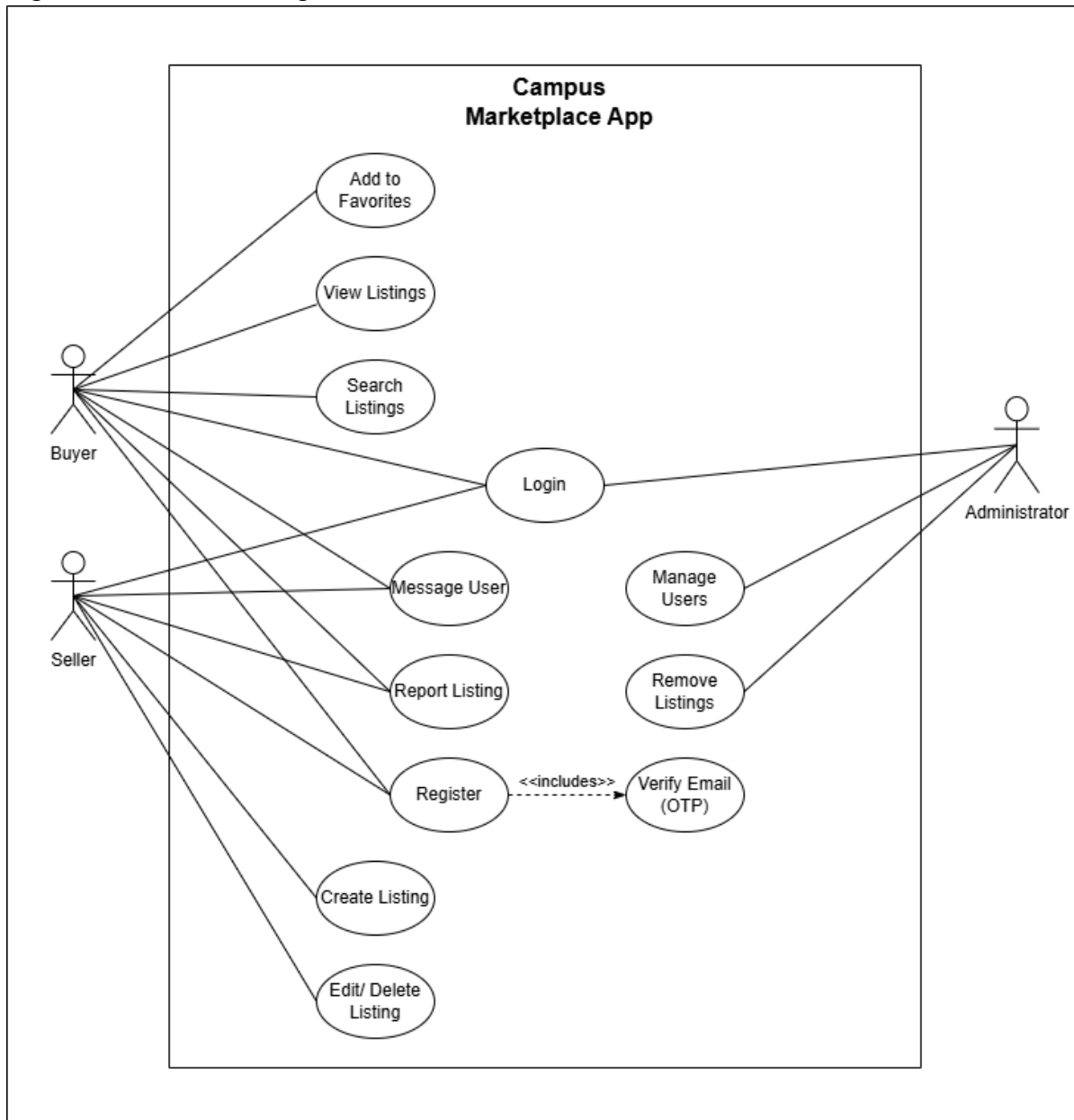
- The system will be compatible with web browsers such as Chrome, Safari, Firefox, and Microsoft Edge.
- The system will provide a responsive web interface accessible on both desktop and mobile devices.
- The system will use relational database management for data storage.
- The system will support image upload functionality for product listings, with uploaded files stored securely.

4.2 Functional Requirements

- Users will be able to register using a valid KPU email.
- Users will be able to log in and log out securely.
- Users will be able to create, edit, and delete listings.
- Users will be able to search and filter listings.
- Users will be able to view listings and their details.
- Users will be able to send and receive messages.
- Administrators will be able to manage users and listings.

4.3 Use Case Model

Figure 4.3: Use Case Diagram



The Use Case Model provides a high-level view of the interactions between three types of actors and the Campus Marketplace Application system. It illustrates system functionality from a user perspective without showing internal implementation details.

4.3.1 UC-01 User Registration

Author: Varleen

Purpose

The purpose of this use case is to allow new users (buyers and sellers) to create an account in the system using a valid institutional email and OTP verification

Requirements Traceability

- User Account Creation
- Email Validation System
- OTP Verification

Priority

- High

Preconditions

- User has access to a valid institutional domain email (@student.kpu.ca, @kpu.ca, e.g.)
- User is not already logged in

Postconditions

- User accounts are successfully created
- Users are authenticated and can access the system

Actors

- Buyer
- Seller

Flow of Events

1. Basic Flow

1. User selects Register option
2. User enters email and password
3. System validates email domain
4. System sends OTP to the user's email
5. User enters the OTP
6. System verifies the OTP
7. System creates the account and logs the user in

2. Alternative Flows

- Invalid Email = If the email domain is not valid → system rejects registration
- Incorrect OTP = If OTP is incorrect → user is prompted to re-enter the OTP
- Missing Fields = If required information is missing → system displays an error

3. Exceptions

- System Failure = registration cannot be completed

Includes Other IDs

- None

Notes/Issues

- Email domain restriction must be enforced
- OTP must be secure

4.3.2 UC-02 User Login

Author: Varleen

Purpose

The login process allows registered users (buyers and sellers) and administrators to access their accounts in the Campus Marketplace Application by entering valid credentials.

Requirements Traceability

- User Authentication
- Account Access Control

Priority

- High

Preconditions

- User has a registered account
- User is not already logged in

Postconditions

- User is authenticated
- User gains access to the system

Actors

- Buyer
- Seller

Flow of events**1. Basic Flow**

1. User selects Login option
2. User enters registered email and password
3. System validates the credentials
4. System grants access to the user

2. Alternative Flows

- Invalid Credentials = If the email or password are not correct → system displays an error message
- Missing Fields = If required fields are empty → system prompts user to complete fields
- Forget Password = If user ask to renew their password → system sends OTP to the user's email.

3. Exceptions

- System error during login

Includes Other IDs

- None

Notes/Issues

- Password security must be enforced

4.3.3 UC-03 Create Listing

Author: Varleen

Purpose

Allows sellers to create and publish listings for items they want to sell on the marketplace.

Requirements Traceability

- Listing Creation
- Product Management

Priority

- High

Preconditions

- Seller is logged into the system

Postconditions

- Listing is created and stored in the system
- Listing becomes visible to other users

Actors

- Seller

Extends

- None

Flow of events**1. Basic Flow**

1. Seller selects Create Listing
2. Seller enters listing details (category, title, price, description, etc.)
3. Seller submits the listing
4. System validates the input
5. System stores the listing in the database
6. Listing is published

2. Alternative Flows

- Missing Information = If required fields are empty → system displays an error message
- Invalid Data = If data format is incorrect → system prompts user to enter data

3. Exceptions

- System error when saving listing

Includes Other IDs

- None

Notes / Issues

- Category selection must be consistent

4.3.4 UC-04 Search Listings

Author: Varleen

Purpose

Allows users to search for available listings using keywords, categories, or filters.

Requirements Traceability

- Search System
- Product Retrieval

Priority

- Medium

Preconditions

- User is logged into the system

Postconditions

- Relevant listings are displayed to the user

Actors

- Buyer
- Seller

Extends

- None

Flow of Events**1. Basic Flow**

1. User enters search keywords or selects filters
2. System processes the search request
3. System retrieves matching listings
4. System displays search results

2. Alternative Flows

- Missing Fields = If required fields are empty → system displays an error message
- No Results Found = If no listings match → system displays “no results found” message.

3. Exceptions

- Database error during search

Includes Other IDs

- None

Notes/Issues

- Search performance should be optimized, user will be notified and asked to try again later

4.3.5 UC-05 Message User

Author: Varleen

Purpose

Allows sellers and buyers to communicate with each other one-to-one regarding a listing through the messaging system.

Requirements Traceability

- Messaging System
- User Communication
- Notification Handling

Priority

- Medium

Preconditions

- User is logged into the system
- Listing exists

Postconditions

- Message is sent and stored in the system
- Recipient can view the message

Actors

- Buyer
- Seller

Extends

- None

Flow of Events

1. Basic Flow

1. User selects a listing
2. User selects Contact Seller/Message User
3. User types a message
4. User sends the message
5. System stores the message
6. Recipient receives the message

2. Alternative Flow

- Message Failure = If message fails → system displays an error
- Empty Message = If no content is entered → system prompts user to enter text

3. Exception

- System errors occur during message processing

Includes Other IDs

- None

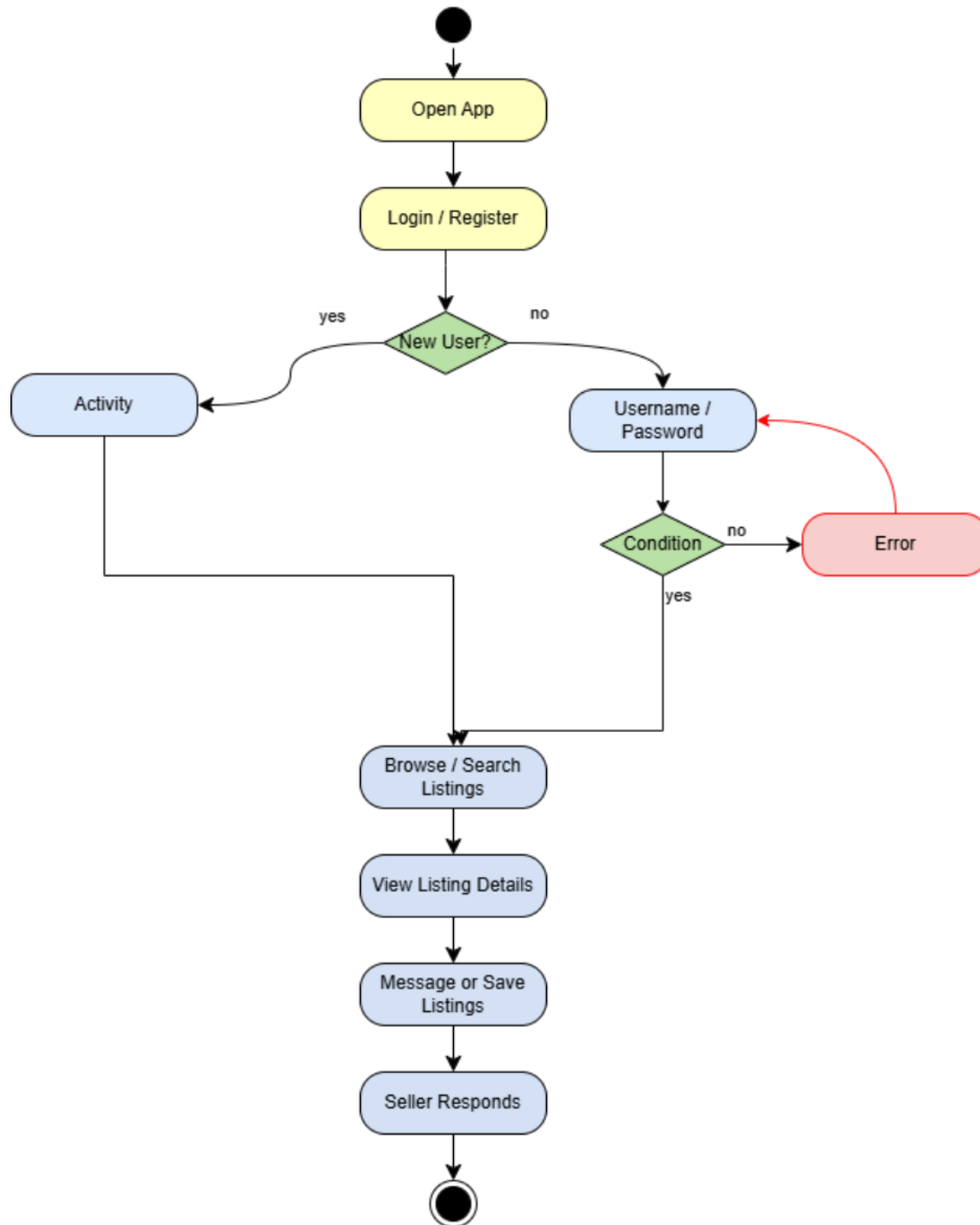
Notes/Issues

- Messaging must support two-way communication between users

4.4 Activity Diagrams

4.4.1 Overall Business Process

Figure 4.4.1: Activity Diagrams

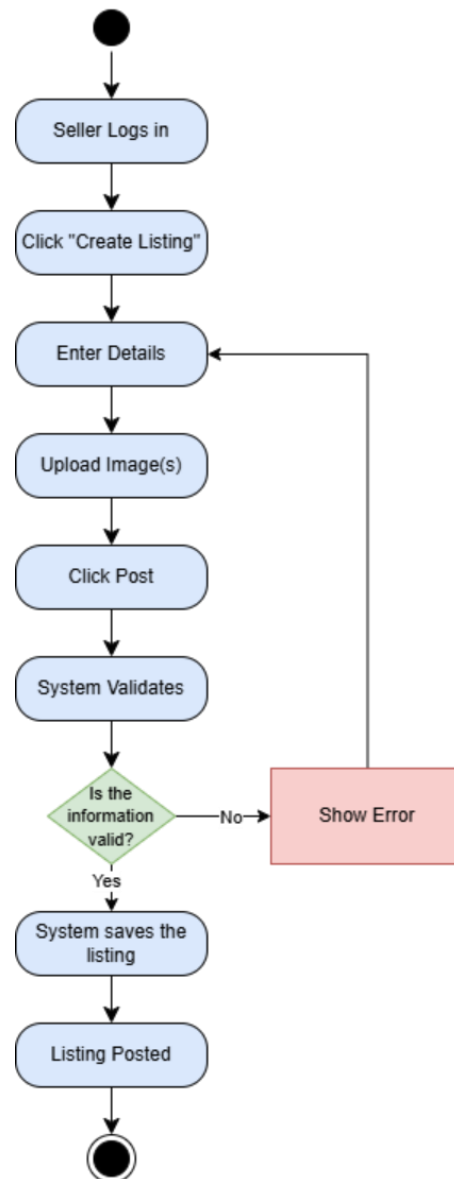


This activity diagram illustrates the workflow of the Compass Marketplace App. It begins when

a user opens the application. A decision point is used to determine whether the user is a new or existing user, leading to either the registration process or login process. After authentication, the user can browse and search listings and view product details. The process continues as users negotiate with sellers. The activity ends when the buyer and seller complete communication and proceed with an offline transaction.

4.4.2 Create Listing Activity Diagram

Figure 4.4.2: Create Listing Activity Diagram

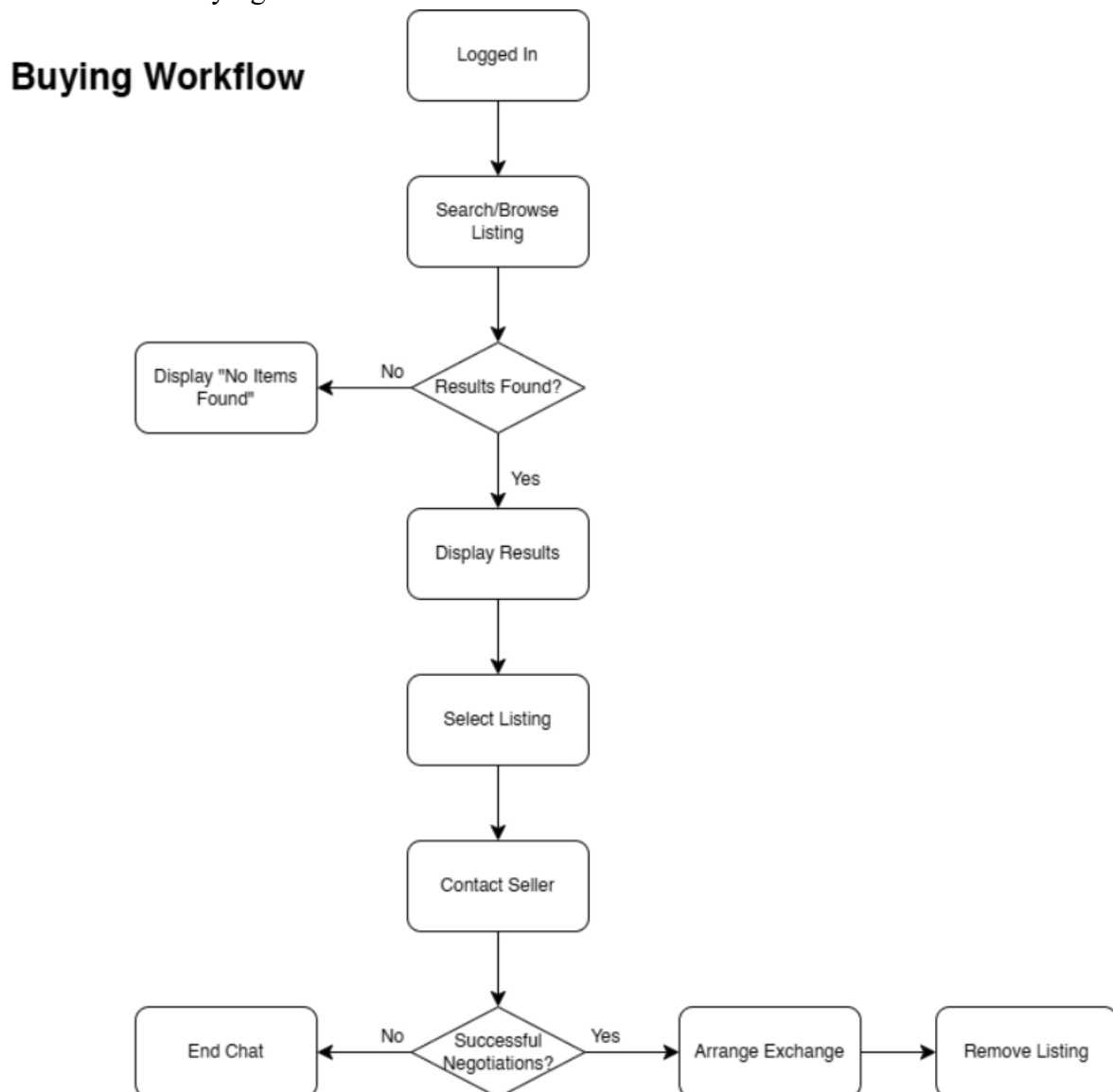


This activity diagram illustrates the workflow of the Create Listings use case. It begins when a

seller clicks the "Create Listing" button. The seller is required to enter product details, including the product name, description, and at least one image. A decision point is used to determine whether the entered information is complete and valid. After validation, the system saves the listing information. The activity ends when the system publishes the new listing to the marketplace.

4.5 Flowcharts

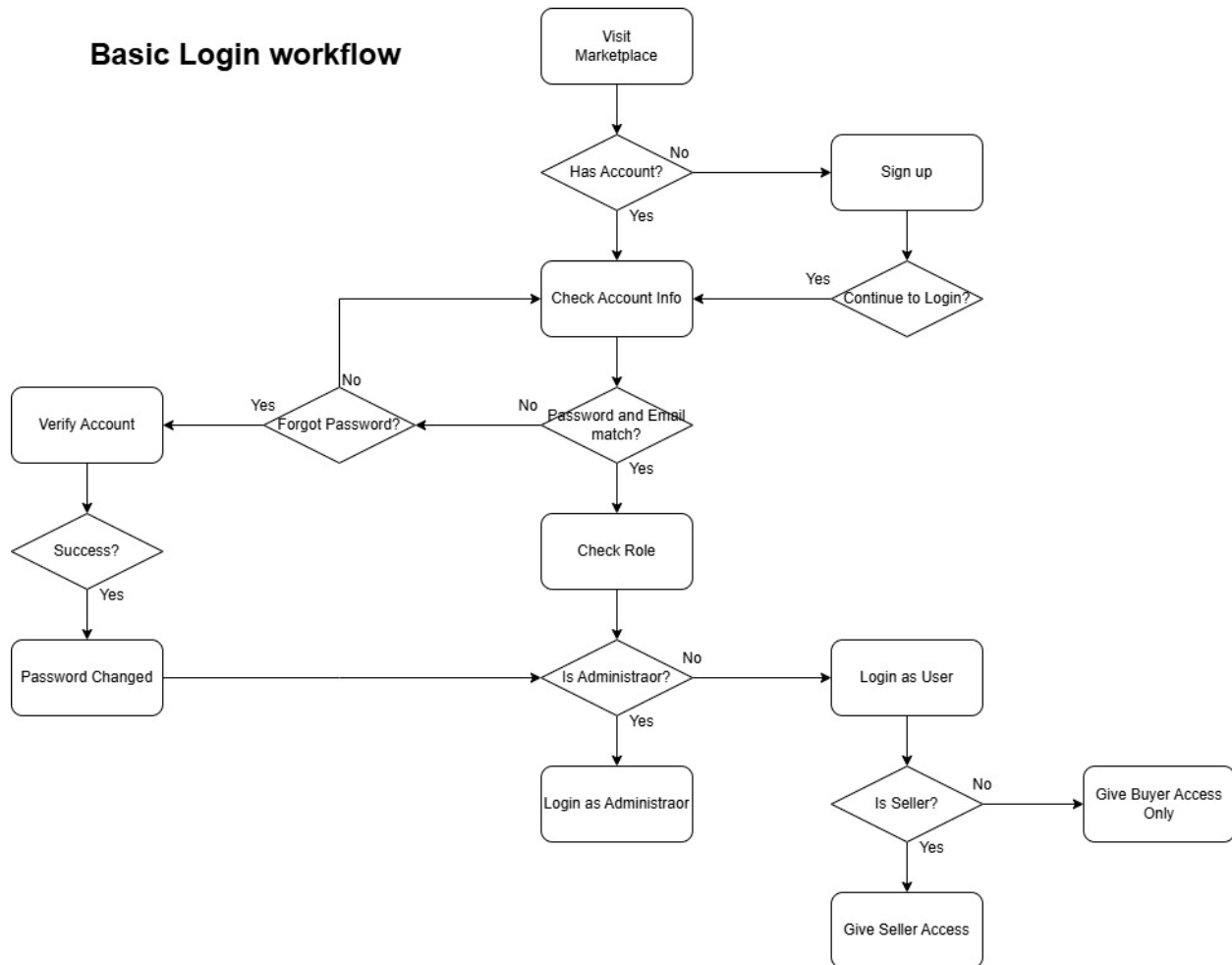
Figure 4.5.1: Basic Buying Flowchart



This flowchart illustrates the buying process of the application. It begins when the user logs in and submits a search request. The system checks for matching listings and displays results if

available. The user can view product details and contact the seller to negotiate. If the negotiation is successful, both parties arrange an exchange, and the process ends with the item being removed from the listing.

Figure 4.5.2: Basic Login Flowchart

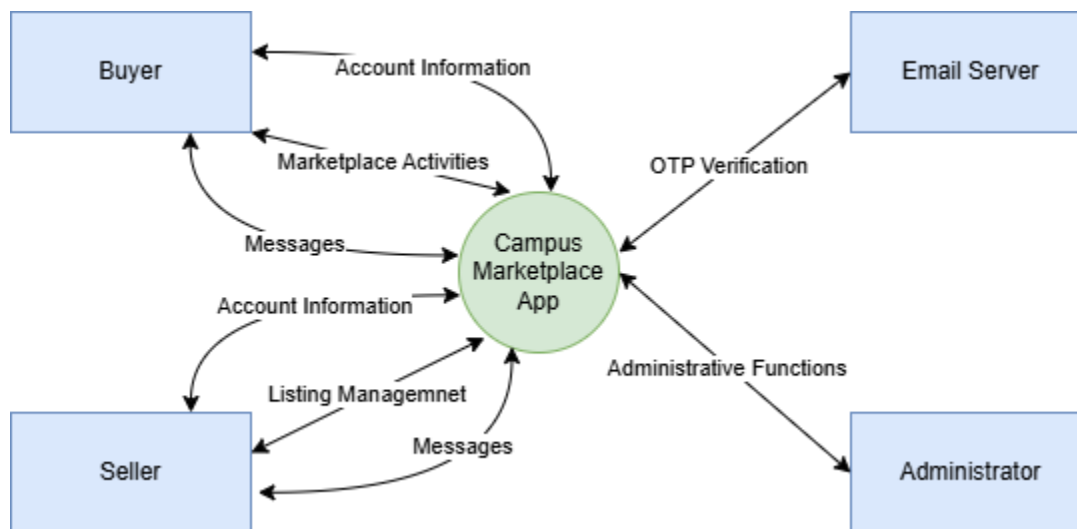


This flowchart illustrates the login process. It begins when the user opens the app. The system first checks whether the account is registered, then validates the username and password. After successful authentication, the system performs role-based authorization to determine whether the user is an administrator, seller, or buyer. The user is then directed to the corresponding interface with appropriate permissions. The process ends when the user is successfully authenticated and logged in.

4.6 Data Flow Diagrams

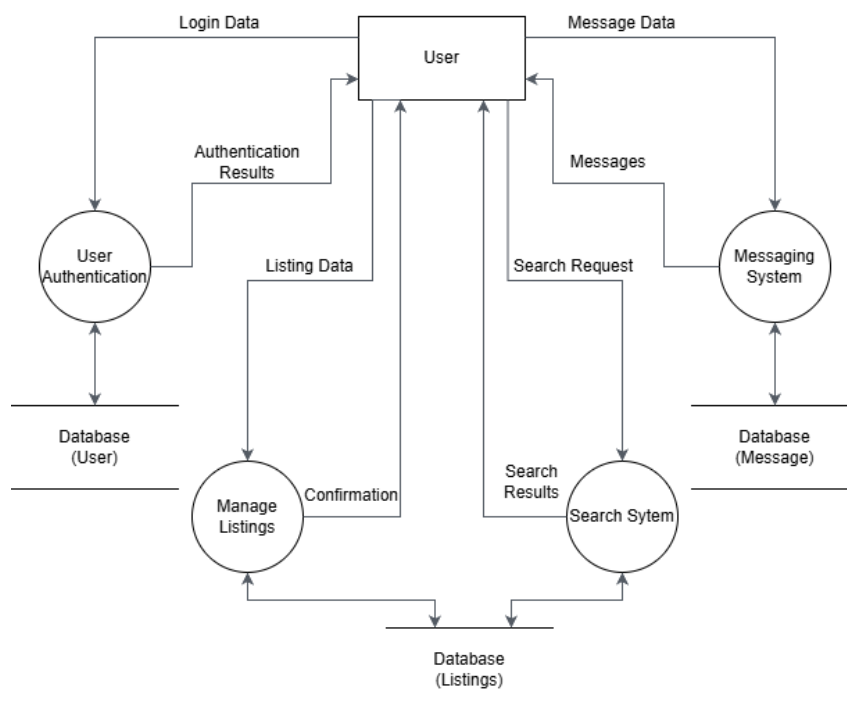
4.6.1 Context Diagram

Figure 4.6.1: Context Diagram



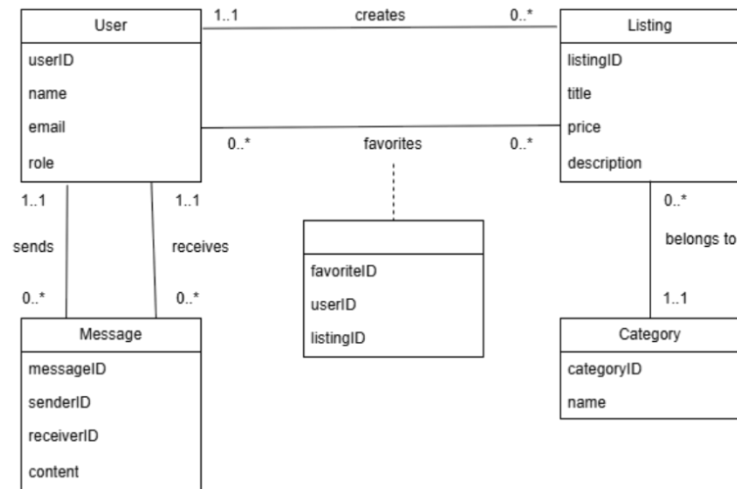
4.6.2 Level 0 Data Flow Diagram

Figure 4.6.2: Data Flow Diagram – Level 0



4.7 Class Diagram

Figure 4.7: Class Diagram



5. Non-Functional Requirements

5.1 Performance Requirements

The system will provide fast and efficient performance to ensure a positive user experience.

- The application will load pages within or less than 3 seconds under normal network conditions.
- The system will support a minimum of 200 concurrent users without losing significant performance of the application.
- Search results will be displayed within or less than 2.5 seconds of a user's query.

5.2 Safety and Security Requirements

The system will protect user information and provide a secure environment for marketplace transactions.

- Users will have to authenticate using their official university email addresses.
- User passwords must meet specific criteria, including a minimum length of eight characters, at least one number, and a special character (\$, %, &, @).
- User passwords will be encrypted and securely stored.
- Only authorized users will be able to access marketplace features.
- Users will be able to report suspicious listings, inappropriate content, or fraudulent activities.

- Personal information will not be shared with third parties without user consent.
- The system will follow and comply with privacy regulations, policies and laws.

5.3 Software Quality Attributes

The system will demonstrate high standards of quality in terms of usability, reliability, and scalability.

5.3.1: Usability

- The user interface will be simple, intuitive, and easy to navigate.
- New users will be able to learn the basic functions of the application without formal training.

5.3.2: Reliability

- The system will operate consistently with minimal errors or failures.

5.3.3: Scalability

- The system will support future growth in the number of users, listings and features without requiring major redesign.

5.3.4: Availability

- The application will be accessible 24 hours a day, except for the scheduled maintenance periods.

5.3.5: Compatibility

- The system will function correctly on major web browsers such as Chrome, Safari, Microsoft Edge, and Firefox.
- The application will be responsive and usable on desktops, tablets, and smartphones.

Appendix A – Data Directory

Table 2: Data Directory

Name	Type	Description	Possible Values	Related Use Case
userID	State Variable	Unique identifier for each user	Numeric (e.g., 1001)	All
name	Input	User's full name	Text	Registration
email	Input	User's email address	Must be institutional email	Registration/Login
password	Input	User's account password	Text	Registration/Login
role	State Variable	Defines type of user	Buyer, Seller, Administrator	All
otp	State Variable	One-time verification code	Numeric (e.g., 6-digit code)	Registration
loginStatus	State Variable	Tracks if user is logged in	Logged In, Logged Out	Login
listingID	State Variable	Unique ID for listing	Numeric	Listings
title	Input	Name of item listed	Text	Create Listing
description	Input	Details about the item	Text	Create Listing
price	Input	Cost of item	Numeric	Create Listing
category	State Variable	Item category	Predefined categories	Create Listing
searchQuery	Input	User search	Text	Search Listings

		keywords		
searchResults	Output	Results from search	List of listings	Search Listings
messageContent	Input	Message body	Text	Message User
messageID	State Variable	Unique ID for message	Numeric	Message User
messageStatus	State Variable	Status of message	Sent, Delivered, Failed	Message User
favorites	State Variable	List of saved items	Collection of listingIDs	Listings
authResult	Output	Login/registration result	Success, Failure	Login/Registration
confirmationMessage	Output	Action confirmation	Success/Failure message	Listings

Appendix B – Interview Reports

The survey was conducted among students and faculty, especially members of the KPU community. To recruit participants, the survey was promoted through KPU student online group chats, physical posters placed on campus, and announcements made to classmates and instructors in our courses.

A total of 31 survey responses were collected, satisfying the minimum number of responses required for feedback collection. The survey data have been analyzed, and the results are presented in Section 3.3. Our survey can be found [here](#).

Appendix C – Group Log

Table 3: Summary Activity Log

Date	Activity	Members Involved
May 25, 2026	Created initial survey questions and discussed required information for the app	Navneet
May 28, 2026	Collected initial survey responses (5 responses)	All Members
June 1, 2026	Created and distributed QR code poster; updated survey questions and expanded audience	Navneet, Varleen
June 2, 2026	Sent emails to previous respondents; created GitHub repository; increased responses (26)	Varleen, Carline
June 3, 2026	Created requirements document structure; discussed listing category feature	Joseph, Varleen, Carline
June 4, 2026	Reviewed rubric; assigned diagram tasks; began presentation; updated document collaboratively	All Members
June 5, 2026	Converted survey data into charts for documentation and presentation	Natash, Navneet
June 6-7, 2026	Completed all diagrams and finalized requirements analysis document	All Members

The group collaborated throughout the project to complete all phases of the requirements analysis for the Campus Marketplace Application. The initial stage involved designing survey questions to gather user requirements, with some team members meeting in person to develop the survey.

The survey was then distributed by all members, and additional efforts such as QR code posters and email outreach were used to increase participation and collect more responses. Most group communication and coordination were conducted through text messaging, allowing for consistent discussion, decision-making, and progress updates with day-to-day notes being made on what was discussed.

As the project progressed, team members worked together to analyze survey data, define system requirements, and determine key system features such as listing categories and user interactions. Tasks were divided among team members to develop various system models, including context diagrams, use case diagrams, class diagrams, and data flow diagrams.

The group also collaborated on creating and refining the Requirements Analysis Document. Each phase of the project was reviewed collectively to ensure accuracy, completeness, and alignment with project objectives. Through continuous collaboration and shared responsibilities, the group successfully completed all required components of the project.